

G MADHUSUDHANA

DEPUTY MANAGER – CNC MACHINE & FIXTURE DESIGN

18+ Years of Engineering Leadership Excellence

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PROFESSIONAL SUMMARY

Mechanical Design Engineer with 18+ years of hands-on experience in CNC machine design, fixture engineering, and new product development. Skilled in leading cross-functional teams, executing automation projects, and delivering reliable, cost-effective machining solutions. Experienced in managing end-to-end design lifecycles for CNC machining centres, fixture systems, and cutting tools. Comfortable with techno-commercial activities, finding new sources, and coordinating between R&D and manufacturing operations.

CORE COMPETENCIES

✓ CNC Machine Design & Development	✓ Fixture Design (Hydraulic / Pneumatic / Modular)	✓ Cutting Tools Design & Optimization
✓ New Product Development (NPD) & R&D	✓ Project Planning, Scheduling & Execution	✓ Team Leadership & Mentorship (10+ Engineers)
✓ FEA / Stress & Deformation Analysis	✓ Process Optimization & Quality Control	✓ Automation – Robot Interfaces & Tool Changers
✓ ERP (SAP S4Hana) & CAD Systems (SolidWorks, CREO, UG)	✓ Techno-Commercial Proposals & Customer Engagement	✓ Finding New Sources & Cost Reduction Initiatives

TECHNICAL SKILLS

CAD Tools	SolidWorks, CREO (Pro-E), Unigraphics (NX), Solid Edge, AutoCAD
CAE / Simulation	SolidWorks Simulation (FEA – Stress, Deformation, Modal Analysis)
CAM Programming	MasterCAM (Basic Level)
ERP Systems	SAP S4Hana, Profit Plus
PDM / Workflow	Autodesk Vault, Teamcenter
Office Tools	MS Office (Word, Excel, PowerPoint, Project)

PROFESSIONAL EXPERIENCE

Ace Designers Limited – MCD (AMS) | Bengaluru

Deputy Manager – Design & Development Oct 2024 – Present

- Handle design of machining fixtures, sub-systems, and tooling for CNC turning and machining centres from concept to release
- Supervise and guide a team of 10 engineers, coordinating project tasks to meet schedules and quality requirements
- Develop and present techno-commercial proposals; conduct detailed customer technical discussions and requirement analysis

- Handle automation projects including robot-machine interfaces, servo-driven pallet changers, and 40-tool auto tool changers
- Perform FEA-based stress, deformation, and rigidity analysis to validate fixture and machine sub-system designs
- Work on part standardization to help reduce cost and inventory complexity across product lines
- Coordinate with manufacturing, assembly, purchase, and service teams to support smooth project execution
- Represent the company at international trade exhibitions: JIMTOF (Tokyo) and TIMTOS (Taipei)

Ace Manufacturing Systems Ltd. | Bengaluru

Senior Engineer / Executive – Fixture Design Division Jan 2005 – Apr 2010

- Designed hydraulic and pneumatic machining fixtures for a wide range of automotive and industrial components
- Developed detailed fabrication, casting, and machining drawings with full GD&T compliance
- Led cross-functional coordination across design, manufacturing, and quality to ensure on-time project delivery
- Designed modular and dedicated fixtures for high-volume production lines at Tier-1 automotive suppliers
- Contributed to cost-reduction initiatives through design-for-manufacturability (DFM) improvements

Timken Engineering & Research – India Pvt Ltd. | Bengaluru

Associate – Design, Ball Product Division May 2010 – Aug 2011

- Designed and modified ball bearing components for global product lines in collaboration with US-based plants
- Delivered precise CAD models and technical documentation using CATIA, Unigraphics, Pro-E, and AutoCAD
- Supported international engineering teams with design reviews, change management, and component validation

PRECI FIT India | Bengaluru

Shift In-Charge – CNC Production Line Jan 2004 – Dec 2004

- Planned and managed CNC production schedules; generated and optimized CNC machining programs
- Developed manual fixtures for aerospace components requiring close dimensional tolerances

Karnataka CNC Tech Pvt Ltd. | Bengaluru

CNC Operator – Vertical Machining Centre Jan 2002 – Dec 2003

- Operated VMC machines for precision component machining, gaining hands-on CNC process knowledge

KEY PROJECTS & ACHIEVEMENTS

- Servo-Driven Linear Automatic Pallet Changer – designed complete mechanism for automated pallet exchange on CNC machining centres
- Twin-Spindle & Auxiliary Spindle Development – engineered multi-spindle configurations for productivity enhancement
- 40-Tool Automatic Tool Changer Implementation – designed and validated ATC system for CNC machining centres
- Brake Caliper Housing Fixture – custom hydraulic fixture for Brakes India Ltd., achieving cycle time reduction
- Axle Fixture – precision machining fixture for Axle India Ltd., meeting tight geometric tolerances
- Multi-Machine Automation – designed and implemented loading/unloading automation across three CNC machines
- International Representation – represented the company at JIMTOF (Tokyo) and TIMTOS (Taipei)

EDUCATION

Bachelor of Technology – Mechanical Engineering

IGNOU – JSSATE, Bangalore

PERSONAL DETAILS

Date of Birth	13 October 1982
Languages	Kannada, English, Hindi, Tamil, Telugu
Marital Status	Married
Nationality	Indian